

- 3 (a) product of (magnitude of one) force and distance between forces M1
 reference to *either* perpendicular distance between forces
 or line of action of forces and perpendicular distance A1 [2]
- (b) (i) 90° B1 [1]
- (ii) $130 = F \times 0.45$ (*allow e.c.f. for angle in (i)*) C1
 $F = 290 \text{ N}$ A1 [2]
 (*allow 1 mark only if angle stated in (i) is not used in (ii)*)

First variant Mark Scheme

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4 (a) (i) change of shape / size / length / dimension

C1